Soft Suave
A KiwiTech Affiliate Company

CASE STUDY · TRANSPORTATION · VISION AI

INTELLIGENT TRANSPORTATION · REAL-TIME VISION AI PLATFORM

# Vehicle Intelligence

## A Vision AI Platform for Real-Time Axle Counting & Traffic Monitoring

Turning top-view highway camera feeds into accurate axle counts, lane-level vehicle tracking and live congestion insight — with specialised recognition of lifted and floating axles on commercial vehicles, deployed in production across multiple high-traffic sites along a busy highway corridor.

| INDUSTRY | SCOPE | ENGAGEMENT |
| --- | --- | --- |
| Intelligent transportation / traffic monitoring (AI) | Real-time axle counting, multi-lane tracking, queue analytics | Production deployment across multiple high-traffic sites |

01 — ABOUT THE CLIENT

# Counting every axle, lane by lane

The client operates traffic-monitoring infrastructure along a busy Mexican highway corridor and set out to build a real-time vehicle axle counting solution. The goal: efficiently count axles, track vehicles by lane and measure throughput in real time — with special attention to lifted or floating axles — while optimising queue counting and top-view processing. And the system had to hold that accuracy in dense, congested traffic, where vehicles overlap in the top-view camera feed.

02 — THE CHALLENGE

# Four problems in real-world traffic

### Lifted & floating axles

Commercial vehicles raise axles clear of the road — traditional counting systems frequently miscounted these lifted axles or missed them entirely.

### Complex vehicle configurations

Real-world traffic mixes many vehicle classes and trailer setups that strain rule-based counting and demand a robust AI approach.

### Real-time, multi-lane load

Queue detection and classification had to run across several lanes simultaneously at near real-time speed to be useful.

### Top-view density & occlusion

In dense, congested traffic vehicles overlap in the top-view camera feed, making clean per-vehicle axle separation far harder.

03 — THE SOLUTION

# One vision pipeline, from detection to insight

Soft Suave built the system as a single computer-vision pipeline over top-view camera feeds — real-time detection, lane-aware tracking, specialised axle recognition and queue analytics in one platform, deployed in production across multiple high-traffic sites and tuned to hold accuracy in dense, congested traffic.

## 01 Detection & Axle Recognition

DETECTION · CLASSIFICATION

Advanced computer-vision algorithms process top-view feeds to count vehicle axles with high precision, even in dense traffic — with a specialised module identifying lifted or floating axles for complete classification accuracy.

| Detection & Axle… | |
|---|---|
| Axles counted | 6 |
| Lifted axle | Detected |
| Confidence | High |

## 02 Lane-Specific Tracking

MULTI-LANE

A sophisticated tracker maintains each vehicle's identity across frames and ties it to its lane, delivering accurate per-lane throughput measurement in real time.

| Lane-Specific Tr… | |
|---|---|
| Lane 3 | 42 veh/hr |
| Track ID | #1147 stable |
| Throughput | Live |

## 03 Queue Analysis Module

CONGESTION

Optimised queue-counting algorithms detect congestion and vehicle accumulation as they build, giving operators the insight needed for proactive traffic responses.

| Queue Analysis M… | |
|---|---|
| Queue length | 14 veh |
| Congestion | Building |
| Alert | Lane 2 |

**Every axle counted, every lane tracked, every queue seen early — one production vision pipeline turning live camera feeds into traffic intelligence.**

04 — KEY FEATURES

# Twelve capabilities, one platform

- Real-Time Axle Counting
- Floating & Lifted Axle Detection
- AI-Powered Vehicle Classification
- Top-View Camera Processing
- Multi-Lane Vehicle Tracking
- Vehicle Throughput Measurement
- Queue & Congestion Analysis
- Proactive Congestion Alerts
- Predictive Congestion Modelling
- Real-Time Traffic Data Feeds
- Enhanced Accuracy in Dense Traffic
- Production Deployment Across Multiple Sites

05 — RESULTS & IMPACT

# Measurable gains across the board

**90%**
Enhanced detection accuracy over the previous monitoring baseline

**80%**
Improvement in vehicle tracking and traffic decision-making

**70%**
Better handling of complex vehicle and trailer configurations

**50%**
Faster access to real-time traffic data

*Figures reflect measured improvements over the client's previous monitoring baseline.*

WHY SOFT SUAVE

## Production-ready intelligence, proven on the road

The client needed more than detection — a real-time vision pipeline handling lifted and floating axles, dense multi-lane traffic and live queue analytics, deployed across multiple high-traffic locations. Soft Suave delivered a scalable, production-ready platform with proven ROI and a future-proof architecture — accuracy that holds in the densest congested traffic, with every lane, vehicle and axle accounted for in real time.

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Let's build what's next.